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PAPER

Inverse engineering for robust state transport along a spin chain via low-energy subspaces

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Abstract

Quantum state transfer (QST) plays a central role in the field of quantum computation and communication, but its quality will be deteriorated by the ubiquitous variations and noise in quantum systems. Here we propose robust and nonadiabatic protocols for transmitting quantum state across a strongly coupled spin chain, especially in the presence of unwanted disorders in the couplings. To this end, we approximately map the low-energy subspaces of the odd-size Heisenberg chain to a two-level system, and derive the sensitivity of the final fidelity with respect to systematic deviations or time-varying fluctuations. Subsequently, leveraging the flexibility of the inverse-engineering technique, we optimize the state-transfer robustness concerning these perturbations individually. The resulting schemes allow for more stable QST than the original accelerated schemes and only require manipulating the two boundary couplings instead of the whole system, which open up the possibility of fast and robust information transfer in spin-based quantum systems.

1. Introduction

To tap the full potentials of quantum information processing and computation, high-fidelity quantum state transfer (QST) between remote qubits is required [1, 2]. Spin chains are paradigmatic systems for short-distance QST due to native spin–spin interactions [3, 4], which can be realized in a variety of physical systems such as Josephson junction arrays [5], quantum dots [6], trapped ions [7] and compounds [8]. Impressive protocols for spin chains typically require to engineer couplings among qubits and evolution time precisely [9–21], which are vulnerable to environmental noises, time errors and etc. On the other hand, the quantum adiabatic control for QST [22–29], though carrying the advantage of inherent robustness to parameter variations, is often time-consuming, and thus it may suffer from decoherence. To mitigate these drawbacks, recently developed schemes [30–39] by shortcut to adiabaticity (STA) [40, 41] have been used to realize QST in arrays of spin qubits, which can greatly accelerate quantum-information processing and simultaneously inherit the robustness of adiabatic dynamics.

However, in reality it is not easy to create identical qubits and ideal quantum information processors based on artificial structures, such as superconducting circuits [42], optical lattices [43] and quantum dots [28]. The emergence of the unwanted disorders from the couplings or external fields in spin chain [44, 45] will spoil the STA passages. Take spin qubits based on electrons in quantum dots for example, an appealing platform for quantum information possessing merits of perceived scalability and long coherence times. The imperfect fabrications and the inevitable gate voltage fluctuations [25–27] would induce perturbations in exchange couplings, and thus the state transport become imperfect. In short, deviations

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from ideal values for various control parameters are inevitable in practical quantum systems [46–53]. It is of importance to engineer more stable shortcuts to QST with various perturbations considered. For instance, [51] proposed a systematic approach to design robust control protocols capable of mitigating the influence of various noise types. By exploiting the traditional control techniques with differentiable programming, Coopmans also investigated the optimal strategies for the robust magnon transport in the presence of disordered magnetic fields [52].

Inspired by these work, here we put forward robust schemes to achieve nonadiabatic quantum state transport along an odd-size spin chain, where the systematic errors and time-varying noise in the qubit-chain couplings are taken into consideration. Both protocols only require the control of the couplings at the edges of the system, which can greatly benefit experimental implementations. Furthermore, they exhibit excellent robustness against undesired perturbations. The rest of the paper is organized as follows. In section 2, we briefly derive the two-level effective quantum system and then present the basic schemes for fast QST based on inverse engineering. Using the perturbation theory, in section 3 a detailed study of the sensitivity on the systematic errors and time-varying noise is provided separately. Then the strategies to mitigate the impact of different perturbations are proposed. Numerical simulations demonstrate the validity of our protocols. Finally, a brief summary and outlook are presented in section 4.

2. Effective two-level Hamiltonian for QST

The Heisenberg spin model is ubiquitous in many physical systems and plays an important role in both condensed-matter physics and quantum computation [8, 9, 28, 43]. The system we consider, two external spins weakly coupled to an odd-size spin chain, is shown in figure 1, and its Hamiltonian is

$$H_0(t) = J_1(t)\,\vec{\sigma}_{\rm S}\cdot\vec{\sigma}_1 + J_2(t)\,\vec{\sigma}_N\cdot\vec{\sigma}_{\rm R} + J_0\sum_{k=1}^{N-1}\vec{\sigma}_k\cdot\vec{\sigma}_{k+1}. \tag{1}$$

Here $\vec{\sigma}=[\sigma^x,\sigma^y,\sigma^z]$ denote Pauli matrices, and the couplings at both ends are much weaker than internal ones, i.e, $J_{1,2}(t)\ll J_0$. As shown in [36, 39], an arbitrary quantum state of the sender $|\psi_0\rangle = a|0\rangle + b|1\rangle$ ($|a|^2+|b|^2=1$) can be transferred to the receiver by initializing the whole system

$$|\Psi(0)\rangle = |\psi_0\rangle\otimes|\psi_{\rm g}\rangle, \tag{2}$$

and then adiabatically switching on the leftmost coupling $J_1(t)$ and off the rightmost one $J_2(t)$. Here $|\psi_{\rm g}\rangle$ is the ground state of the rest subsystems. The target state is $|\Psi(T)\rangle = |\psi_{\rm g}\rangle\otimes|\psi_0\rangle$, and the final QST quality can be assessed by performing quantum state tomography on the receiver.

The odd-size spin channel has twofold degenerate ground states $\{|0_{\rm C}\rangle, |1_{\rm C}\rangle\}$, and it behaves like a central spin at the low energies when two qubits are weakly coupled to this chain [12, 54, 55]. The effective Hamiltonian to first order is

$$H_1(t) = \tilde{J}_1(t)\,\vec{\sigma}_{\rm S}\cdot\vec{\sigma}_{\rm C} + \tilde{J}_2(t)\,\vec{\sigma}_{\rm C}\cdot\vec{\sigma}_{\rm R}, \tag{3}$$

where the central spin operator $\vec{\sigma}_{\rm C}$ is defined by $\sigma^x_{\rm C} = |0_{\rm C}\rangle\langle 1_{\rm C}| + |1_{\rm C}\rangle\langle 0_{\rm C}|$, $\sigma^y_{\rm C} = -i|0_{\rm C}\rangle\langle 1_{\rm C}| + i|1_{\rm C}\rangle\langle 0_{\rm C}|$, $\sigma^z_{\rm C} = |0_{\rm C}\rangle\langle 0_{\rm C}| - |1_{\rm C}\rangle\langle 1_{\rm C}|$, and the effective couplings $\tilde{J}_{1,2}$ read

$$\tilde{J}_1(t) = mJ_1(t), \quad \tilde{J}_2(t) = mJ_2(t). \tag{4}$$

Here $m = m_{1,N} = \langle 0_{\rm C}|\sigma^z_{1,N}|0_{\rm C}\rangle$ represent the dimensionless local magnetic moments of the first and end spins of the chain [12]. To simplify notations, we use subscripts $\{1,2,3\}$ to represent the three spins in the following. Thus, the initial and target states become $|\tilde{\Psi}(0)\rangle = (a|0\rangle + b|1\rangle)\otimes(|01\rangle - |10\rangle)/\sqrt{2}$ and $|\tilde{\Psi}(T)\rangle = (|01\rangle - |10\rangle)\otimes(a|0\rangle + b|1\rangle)/\sqrt{2}$.

Notice that the effective system preserves the number of excitations during the evolution for $[H_1, \sum_{i=1}^3\sigma_i^z] = 0$, which enables H_1 to be diagonalized in the rearranged basis $\{|000\rangle, |100\rangle, |010\rangle, |001\rangle, |011\rangle, |101\rangle, |110\rangle, |111\rangle\}$, that is,

$$H^{\mathcal{X}} = {\rm diag}\left\{H_1^{\mathcal{X}_+}, H_2^{\mathcal{X}_+}, H_2^{\mathcal{X}_-}, H_1^{\mathcal{X}_-}\right\}. \tag{5}$$



Figure 1. Schematic of QST through an odd-size Heisenberg spin chain, where two qubits S and R are weakly coupled to the channel.

Here $H_1^{\mathcal{X}_\pm}=\tilde{J}_1+\tilde{J}_2$ in the subspaces $\mathcal{X}_1^+=\{|000\rangle\}$ and $\mathcal{X}_1^-=\{|111\rangle\}$, respectively; $H_2^{\mathcal{X}_\pm}$ take the form of

$$H_2^{\mathcal{X}_\pm}(t)=\begin{bmatrix}\tilde{J}_2-\tilde{J}_1 & 2\tilde{J}_1 & 0\\ 2\tilde{J}_1 & -\tilde{J}_1-\tilde{J}_2 & 2\tilde{J}_2\\ 0 & 2\tilde{J}_2 & \tilde{J}_1-\tilde{J}_2\end{bmatrix} \tag{6}$$

in the subspaces $\mathcal{X}_2^\pm$ spanned by $\{|100\rangle, |010\rangle, |001\rangle\}$ and $\{|011\rangle, |101\rangle, |110\rangle\}$, respectively.

At time $t=0$, the ground states of the system in subspaces $\mathcal{X}_2^\pm$ are $|G_{\mathcal{X}_+}(0)\rangle=(|001\rangle-|010\rangle)/\sqrt{2}$ and $|G_{\mathcal{X}_-}(0)\rangle=(|101\rangle-|110\rangle)/\sqrt{2}$, respectively, so $|\tilde{\Psi}(0)\rangle$ can be expressed as the superposition of the two ground states

$$|\tilde{\Psi}(0)\rangle=a|G_{\mathcal{X}_+}(0)\rangle+b|G_{\mathcal{X}_-}(0)\rangle, \tag{7}$$

which only involves the subspaces $\mathcal{X}_2^+$ and $\mathcal{X}_2^-$. Due to the exciton conservation, the transitions between the subspaces with different numbers of excitons will not occur. The Hamiltonian (6) in both subspaces have the same form, which will generate the same dynamics in $\mathcal{X}_2^\pm$. Thus, we can consider the system dynamics in either subspace, for example, $\mathcal{X}_2^+$. The initial state turns to $|\tilde{\Psi}(0)\rangle=(|001\rangle-|010\rangle)/\sqrt{2}$. Meanwhile, the Hamiltonian (3) can be simplified as $H_2^{\mathcal{X}_+}(t)$, which can be further block diagonalized by a unitary transformation

$$H_v^{\mathcal{X}_+}=V^\dagger H_2^{\mathcal{X}_+}V=\begin{bmatrix}\tilde{J}_1-2\tilde{J}_2 & \sqrt{3}\tilde{J}_2 & 0\\ \sqrt{3}\tilde{J}_2 & -3\tilde{J}_1 & 0\\ 0 & 0 & \tilde{J}_1+\tilde{J}_2\end{bmatrix}. \tag{8}$$

Here $V=\{|\xi_1^+\rangle, |\xi_2^+\rangle, |\xi_3^+\rangle\}$ and the basis are

$$|\xi_1^+\rangle=\frac{1}{\sqrt{6}}\begin{pmatrix}1\\1\\-2\end{pmatrix}, |\xi_2^+\rangle=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\0\end{pmatrix}, |\xi_3^+\rangle=\frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix},$$

in the subspace $\mathcal{X}_2^+=\{|100\rangle, |010\rangle, |001\rangle\}$. It follows that the initial state in this new basis reads

$$|\Psi(0)\rangle_v=V^\dagger|\tilde{\Psi}(0)\rangle=-\frac{\sqrt{3}}{2}|\xi_1^+\rangle+\frac{1}{2}|\xi_2^+\rangle, \tag{9}$$

which only consists of the first two basis. Thus, the system driven by $H_v^{\mathcal{X}_+}$ would remain in the sub-subspaces spanned by $\{|\xi_1^+\rangle, |\xi_2^+\rangle\}$. The Hamiltonian (8) can be mapped into a two-level system and eventually simplified as

$$\begin{aligned}\mathcal{H}_0&=\begin{bmatrix}\tilde{J}_1-2\tilde{J}_2 & \sqrt{3}\tilde{J}_2\\ \sqrt{3}\tilde{J}_2 & -3\tilde{J}_1\end{bmatrix}\\ &=\left(2\tilde{J}_1-\tilde{J}_2\right)\sigma^z+\sqrt{3}\tilde{J}_2\sigma^x-\left(\tilde{J}_2+\tilde{J}_1\right)\mathbf{I}_2.\end{aligned} \tag{10}$$

Here $\mathbf{I}_2$ represents the 2×2 identity matrix. The last term contributes only a global phase in the evolution, which can be neglected for simplicity. Therefore, the original QST scheme in the spin chain (1) can be dealt with the effective two-level Hamiltonian in the basis $\{|\xi_1^+\rangle, |\xi_2^+\rangle\}$

$$\mathcal{H}_0=\frac{h_z(t)}{2}\sigma^z+\frac{h_x(t)}{2}\sigma^x \tag{11}$$

with $h_z(t)=4\tilde{J}_1(t)-2\tilde{J}_2(t)$ and $h_x(t)=2\sqrt{3}\tilde{J}_2(t)$. The corresponding initial and target states are

$$|\Psi(0)\rangle_v=\frac{1}{2}\begin{pmatrix}-\sqrt{3}\\1\end{pmatrix}, \quad |\Psi(T)\rangle_v=\begin{pmatrix}0\\1\end{pmatrix}. \tag{12}$$

To construct the feasible shortcuts for QST, we can adopt the inverse engineering method. For a two-level system, the dynamical invariant of the Schrödinger equation is given by [56]

$$\mathcal{I}(t)=\frac{\mu}{2}\begin{bmatrix}\cos\theta & \mathrm{e}^{-\mathrm{i}\beta}\sin\theta\\ \mathrm{e}^{\mathrm{i}\beta}\sin\theta & -\cos\theta\end{bmatrix}, \tag{13}$$

satisfying the following relation ($\hbar=1$)

$$\frac{\partial}{\partial t}\mathcal{I}(t)+i[\mathcal{H}_0(t),\mathcal{I}(t)]=0. \tag{14}$$

Here the parameters θ and β are the polar and azimuthal angles in the Bloch sphere, meeting the relationships with $h_{x,z}$:

$$\dot{\theta}=-h_x\sin\beta, \tag{15}$$

$$\dot{\beta}=h_z-h_x\cot\theta\cos\beta, \tag{16}$$

where the dot denotes the time derivative.

The corresponding eigenvectors of the invariant $\mathcal{I}(t)$ are

$$|\phi_+(t)\rangle=\begin{pmatrix}\mathrm{e}^{-\mathrm{i}\beta}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}\end{pmatrix},|\phi_-(t)\rangle=\begin{pmatrix}\sin\frac{\theta}{2}\\ -\mathrm{e}^{\mathrm{i}\beta}\cos\frac{\theta}{2}\end{pmatrix}, \tag{17}$$

with the eigenvalues $\pm\frac{\mu}{2}$. Then the solution of the time-dependent Schrödinger equation can be written as

$$|\Psi(t)\rangle=\sum_n c_n|\phi_n(t)\rangle\,\mathrm{e}^{\mathrm{i}\gamma_n}, \tag{18}$$

in which the Lewis–Riesenfeld phase reads

$$\gamma_\pm=\pm\frac{1}{2}\int_0^t \mathrm{d}t'\left(\dot{\beta}+\frac{\dot{\theta}\cot\beta}{\sin\theta}\right). \tag{19}$$

Finally, we can choose the state evolves along the dynamical mode $|\phi_-(t)\rangle$ for nonadiabatic QST by imposing the boundary conditions

$$\theta(0)=2\pi/3,\quad \theta(T)=0,\quad \beta(0)=0. \tag{20}$$

Furthermore, equations (15) and (16) suggest the following conditions

$$\dot{\theta}(0)=0,\quad \dot{\theta}(T)=0,\quad \beta(T)=\pi/2, \tag{21}$$

should be added, so that the control fields are finite and smooth at the boundaries. Meanwhile, the invariant $\mathcal{I}$ and $\mathcal{H}_0$ commute at the final time, which share the same eigenstates. Once $\theta(t)$ and $\beta(t)$ are interpolated with the boundaries, the time-dependent couplings $J_{1,2}(t)$ can be inversely designed, and theoretically numerous schemes satisfy these boundaries. A major purpose of our work is to improve the state-transfer robustness against different perturbations by virtue of the inverse engineering.

3. Robust nonadiabatic QST

In this section we explore the influence of the disorders in exchange couplings on QST schemes. Note that creating a desirable spin channel with uniform couplings in the quantum system is challenging, and the inner couplings J_0 inevitably have static variations. Nevertheless, the spin channel with the strong couplings still behaves as an effective spin-1/2 system [48]. Therefore, we primarily assume that the disorders take place on the time-dependent $J_{1,2}(t)$ and are independent of the others, which can be simulated as

$$J_i(t)=J_i(t)+\epsilon_i^{\mathrm{sys}}+\epsilon_i^{\mathrm{rand}}(t),\quad i=1,2. \tag{22}$$

Here ϵ^{sys} and $\epsilon^{\mathrm{rand}}(t)$ denote the static systematic error and time-varying fluctuations, respectively. Next the aforementioned two types of perturbations are separately considered.

3.1. Systematic errors

Due to imperfect fabrications and possible impurities, the time-dependent exchange couplings in the spin chain will suffer from the constant systematic errors $\epsilon_i^{\rm sys}$. We here suppose the two systematic errors have relative ratio $\epsilon_2^{\rm sys}=\mathcal{K}\epsilon_1^{\rm sys}$. According to equations (4) and (11), the corresponding perturbative Hamiltonian for these errors can be described by

$$\mathcal{H}'=\delta\left[(2-\mathcal{K})\sigma^z+\sqrt{3}\mathcal{K}\sigma^x\right] \tag{23}$$

with $\delta=m\epsilon_1^{\rm sys}$. Using the time-dependent purturbation theroy up to second order, we have

$$\begin{aligned}|\Psi'(T)\rangle=&|\Phi_-(T)\rangle-i\int_0^T{\rm d}tU_0(T,t)\mathcal{H}'|\Phi_-(t)\rangle\\&-\int_0^T{\rm d}t\int_0^t{\rm d}t'U_0(T,t)\mathcal{H}'U_0(t,t')\mathcal{H}'|\Phi_-(t')\rangle,\end{aligned}$$

where the unperturbed time evolution operator

$$U_0(T,t)=|\Phi_+(T)\rangle\langle\Phi_+(t)|+|\Phi_-(T)\rangle\langle\Phi_-(t)| \tag{24}$$

with $|\Phi_\pm(t)\rangle={\rm e}^{{\rm i}\gamma_\pm(t)}|\phi_\pm(t)\rangle$. Thus, at final time the quantum fidelity with respect to the target state is

$$\mathcal{F}\simeq 1-\left|\int_0^T{\rm d}t\langle\Phi_+(t)|\mathcal{H}'|\Phi_-(t)\rangle\right|^2 \tag{25}$$

$$=1-\delta^2\left|(2-\mathcal{K})\mathcal{M}+\sqrt{3}\mathcal{K}\mathcal{N}\right|^2 \tag{26}$$

with

$$\mathcal{M}=\int_0^T{\rm d}t{\rm e}^{{\rm i}2\gamma_-}{\rm e}^{{\rm i}\beta}\sin\theta, \tag{27}$$

$$\mathcal{N}=\int_0^T{\rm d}t{\rm e}^{{\rm i}2\gamma_-}{\rm e}^{{\rm i}\beta}(-\cos\beta\cos\theta-i\sin\beta). \tag{28}$$

By defining the systematic-error sensitivity (SES) as [56]

$$Q_{\rm s}:=-\frac{1}{2}\frac{\partial^2\mathcal{F}}{\partial\delta^2}\bigg|_{\delta=0}=-\frac{\partial\mathcal{F}}{\partial(\delta^2)}\bigg|_{\delta=0}, \tag{29}$$

the optimally robust invariant-based shortcuts are those making $Q_{\rm s}$ as small as possible. Since it is hard to get the analytical solution here, we utilize numerical method for minimizing $Q_{\rm s}$. Following the references [57, 58], we assume

$$2\gamma_-+\beta=-2\theta-2\alpha\sin2\theta+2\sqrt{\frac{2\pi}{3}-\theta}=\mathcal{Z}(\alpha,\theta). \tag{30}$$

Here $\theta(t)\in[0,2\pi/3]$ and α is a to-be-optimized parameter. Differentiating the above equation, we have

$$2\dot\gamma_-+\dot\beta=\dot\theta\left[-2-4\alpha\cos2\theta-\frac{1}{\sqrt{\frac{2\pi}{3}-\theta}}\right]. \tag{31}$$

From equation (19) we get

$$2\dot\gamma_-+\dot\beta=-\frac{\dot\theta\cot\beta}{\sin\theta}. \tag{32}$$

Then, combining the equations (31) and (32), one has

$$\cot\beta=\left(2+4\alpha\cos2\theta+\frac{1}{\sqrt{\frac{2\pi}{3}-\theta}}\right)\sin\theta=\mathcal{W}(\alpha,\theta). \tag{33}$$

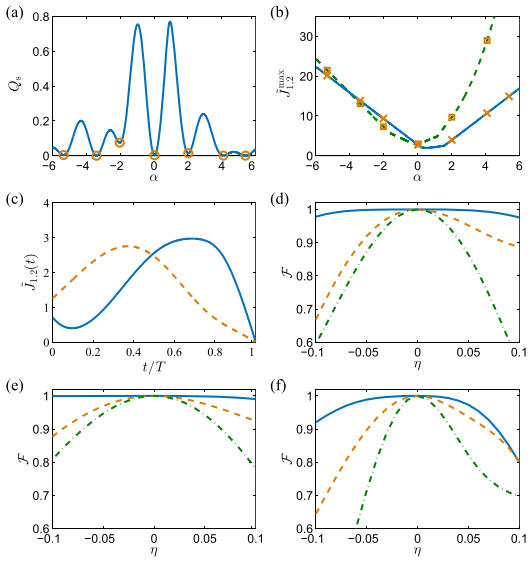


Figure 2. (a) Systematic-error sensitivity Q_s versus the parameter α when $\mathcal{K}=1$. (b) The squares and crosses denote $\tilde{J}_1^{\max}$ and $\tilde{J}_2^{\max}$, respectively, when Q_s is local minimum in subgraph (a). (c) The time-dependent couplings $\tilde{J}_{1,2}(t)$ in our optimal protocol versus t/T for $\alpha_m=0.030$. (d) Fidelity $\mathcal{F}$ as a function of the relative systematic error η where the SES-optimized shortcut (solid line), non-optimized method (dashed line), and the UECD method (dashed-dotted line) are compared. Here we set evolution time $T=1$. (e) and (f) Comparison of the fidelity using three protocols for $\mathcal{K}=0.2$ and -2.

Finally, the SES can be rewritten as

$$Q_s=\left|\int_0^T \mathrm{d}t e^{\mathrm{i}\mathcal{Z}}\left[(2-\mathcal{K})\sin\theta-\frac{\sqrt{3}\mathcal{K}\left(i+\mathcal{W}\cos\theta\right)}{\sqrt{\mathcal{W}^2+1}}\right]\right|^2,$$

and the corresponding couplings $\tilde{J}_{1,2}(t)$ are

$$\tilde{J}_1(t)=-\frac{\dot{\theta}}{4}\left[\frac{\frac{d\mathcal{W}}{d\theta}+\left(\mathcal{W}+\mathcal{W}^3\right)\cot\theta}{1+\mathcal{W}^2}+\sqrt{\frac{\mathcal{W}^2+1}{3}}\right], \tag{34}$$

$$\tilde{J}_2(t)=-\frac{\dot{\theta}}{2}\sqrt{\frac{\mathcal{W}^2+1}{3}}. \tag{35}$$

We are interested in the protocol which the maximum of $\tilde{J}_{1,2}(t)$, denoted by $\tilde{J}_{1,2}^{\max}=\max_{0\leqslant t\leqslant T}\{\tilde{J}_{1,2}(t)\}$, is as small as possible. In other words, a small α is preferable according to equations (34) and (35) if Q_s works similarly for different α. To interpolate at intermediate times, we choose $\theta(t)$ with polynomial function

$$\theta(t)=\frac{2\pi}{3}-\frac{2\pi t^2}{T^2}+\frac{4\pi t^3}{3T^3}. \tag{36}$$

Figure 2(a) shows the SES as a function of α for $\mathcal{K}=1$. The possible solutions to these local minimal Q_s are labeled with circles in the interval $\alpha\in[-6,\ 6]$. The corresponding maximum amplitude of $\tilde{J}_{1,2}(t)$ (squares and crosses), as depicted in figure 2(b), increases monotonously with $|\alpha|$. Therefore the optimal solution here is $\alpha_m=0.030$, which not only makes $Q_s=0.28\times10^{-2}$ almost zero but also provides the smallest $\tilde{J}_{1,2}^{\max}$. Figure 2(c) presents the time-dependent couplings $\tilde{J}_{1,2}(t)$ in this case. Figure 2(d) shows the robustness of our SES-optimized shortcut (solid line) versus the relative systematic error $\eta=\delta/\max\{\tilde{J}_1^{\max},\tilde{J}_2^{\max}\}$. For comparison, we also numerically simulate the results using: (i) the non-optimized method (dashed line) with $\beta(t)=\frac{\pi t}{2T}$ and $\theta(t)$ taking the same form of equation (36), (ii) the unitarily equivalent counterdiabatic (UECD) driving (dashed-dotted line; details can be found in the appendix A). Our simulation manifests the SES-optimized scheme has excellent robustness, even if the relative systematic error $|\eta|=10\%$. Figures 2(e) and (f) present the results for different ratio $\mathcal{K}$ with the same procedure, which also confirm the good stability of the SES-optimized shortcuts.

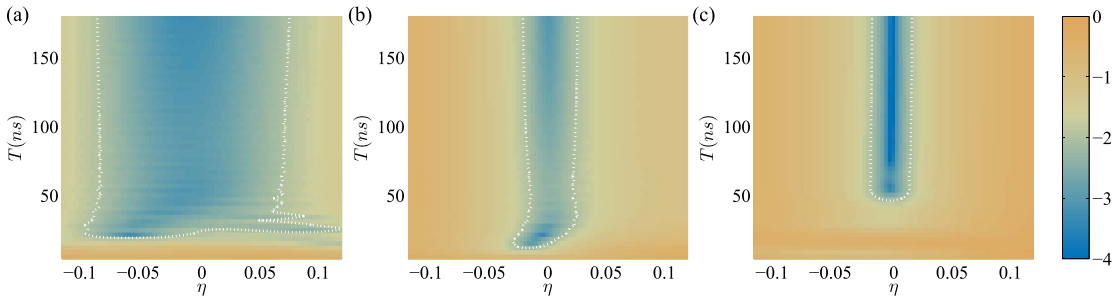


Figure 3. The logarithm of infidelity $\log_{10}(1-F)$ in five-spin chains versus the relative systematic error η using the (a) SES-optimized scheme, (b) non-optimized protocol, and (c) UECD driving when $\mathcal{K}=1$. In all subfigures $J_0 = 120$ MHz and the dashed isolines denote the fidelity being 0.99. The color bar represents the value of $\log_{10}(1-F)$.

We also explore the dynamical evolution in the whole spin chain, by substituting $J_{1,2}(t)$ in original Hamiltonian H_0 with the newly redesigned couplings, and assess the QST fidelity F at the final time T, i.e.

$$F(T) = \langle\psi_0|\varrho_f(T)|\psi_0\rangle, \tag{37}$$

in which $|\psi_0\rangle = a|0\rangle + b|1\rangle$ is the target state and

$$\varrho_f(T) = \mathrm{Tr}_{\overline{\mathrm{R}}}\left[U(0,T)|\Psi(0)\rangle\langle\Psi(0)|U^{\dagger}(0,T)\right] \tag{38}$$

denotes the reduced density matrix of the last spin by tracing over others. Without loss of generality, we carry out the numerical simulations on a five-spin chain. Here the coefficients a, b are randomly chosen as $a = \cos(\frac{\pi}{5})$, $b = \sin(\frac{\pi}{5})e^{-\mathrm{i}\frac{\pi}{4}}$, respectively. Note that the other values have no influence on the final results. Figure 3 presents the QST fidelity F with different methods when we consider the relative systematic error η for $\mathcal{K}=1$. We find that the high-fidelity QST can be realized within several nanoseconds. More importantly, compared to the non-optimized and UECD protocols, the optimized SES scheme is more robust, where F can exceed 0.99 in most situations. Note the inner exchange couplings J_0 is actually finite. If we transport the state as fast as possible i.e. $T \to 0$, the infidelity from imperfectly mapping the whole system to the simplified model would appear obviously in all cases.

3.2. Stochastic amplitude noise

Now we consider the robustness with respect to noise. Specifically, we assume there exists independent noise in two manipulated couplings $J_{1,2}(t)$, which can be expressed as the form $\lambda\xi_i(t)\mathcal{H}_i(t)$ with $i = 1,2$. Here λ and $\mathcal{H}_i(t)$ are the noise strength and the 'noisy' Hamiltonian, respectively; $\xi_i(t)$ denoting a given noise realization satisfies [59]

$$\langle\xi_i(t)\rangle = 0, \quad \langle\xi_i(t)\xi_i(t')\rangle = C(t-t'), \tag{39}$$

where $C(t-t')$ is the correlation function of the noise. According to the ratio between the noise correlation time and the typical system time scale, the corresponding master equations have been derived approximately [60]. In this paper we only consider the amplitude of couplings $J_{1,2}(t)$ with the common Gaussian white noise, which contributes to analytical treatment and understanding more complex noise types. The corresponding correlation function is $C(t-t') = \delta(t-t')$, and the power spectrum is constant. Thus, the master equation ($\hbar = 1$) can be derived [53, 56, 59, 61, 62]:

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho = -i[\mathcal{H}_0,\rho] - \frac{\lambda^2}{2}\left([\mathcal{H}_1,[\mathcal{H}_1,\rho]] + [\mathcal{H}_2,[\mathcal{H}_2,\rho]]\right), \tag{40}$$

where ρ is the density matrix and the operators

$$\mathcal{H}_1 = \begin{bmatrix} 2\tilde{J}_1 & 0 \\ 0 & -2\tilde{J}_1 \end{bmatrix}, \quad \mathcal{H}_2 = \begin{bmatrix} -\tilde{J}_2 & \sqrt{3}\tilde{J}_2 \\ \sqrt{3}\tilde{J}_2 & \tilde{J}_2 \end{bmatrix}. \tag{41}$$

For convenience, we can represent ρ by the Bloch vector

$$\vec{r}=\begin{pmatrix}\rho_{12}+\rho_{21}\\ i(\rho_{12}-\rho_{21})\\ \rho_{11}-\rho_{22}\end{pmatrix} \tag{42}$$

such that $\rho=\frac{1}{2}(1+\vec{r}\cdot\vec{\sigma})$. The Bloch equation for the master equation (40) is

$$\frac{\mathrm{d}}{\mathrm{d}t}\vec{r}=\left[\mathcal{L}_0-\lambda^2(\mathcal{L}_1+\mathcal{L}_2)\right]\vec{r}, \tag{43}$$

in which $\mathcal{L}_{0,1,2}$ denote the operators $\mathcal{H}_{0,1,2}$ in the basis of Pauli operators, respectively, i.e.

$$\mathcal{L}_0=\begin{bmatrix}0 & -h_z & 0\\ h_z & 0 & -h_x\\ 0 & h_x & 0\end{bmatrix},\quad \mathcal{L}_1=\frac{\mathcal{J}_1^2}{2}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix},\quad \mathcal{L}_2=\frac{\mathcal{J}_2^2}{2}\begin{bmatrix}1 & 0 & \sqrt{3}\\ 0 & 4 & 0\\ \sqrt{3} & 0 & 3\end{bmatrix}, \tag{44}$$

with $h_z=4\tilde{J}_1-2\tilde{J}_2, h_x=2\sqrt{3}\tilde{J}_2$, $\mathcal{J}_1=4\tilde{J}_1$, and $\mathcal{J}_2=2\tilde{J}_2$.

The initial and target states can be written as

$$\vec{r}(0)=-\frac{1}{2}\begin{pmatrix}\sqrt{3}\\ 0\\ -1\end{pmatrix},\quad \vec{r}(T)=\begin{pmatrix}0\\ 0\\ -1\end{pmatrix}, \tag{45}$$

respectively. It is known that the unperturbed Bloch vector is

$$\vec{r}_0(t)=\begin{pmatrix}\sin\theta\cos\beta\\ \sin\theta\sin\beta\\ \cos\theta\end{pmatrix}. \tag{46}$$

We can obtain the final fidelity utilizing the time-dependent perturbation theory

$$\mathcal{F}\simeq 1-\frac{\lambda^2}{2}\int_0^T \mathrm{d}t\vec{r}_0(t)^T\left[\mathcal{L}_1+\mathcal{L}_2\right]\vec{r}_0(t). \tag{47}$$

Thus, the amplitude–noise sensitivity (ANS) can be defined as [56]

$$Q_\mathrm{r}:=-\frac{1}{2}\frac{\partial^2\mathcal{F}}{\partial\lambda^2}\bigg|_{\lambda=0}=-\frac{\partial\mathcal{F}}{\partial(\lambda^2)}\bigg|_{\lambda=0}. \tag{48}$$

It indicates that a smaller Q_r can give a more robust scheme for QST. According to equations (47) and (48), we get

$$Q_\mathrm{r}=\int_0^T \mathrm{d}t\mathcal{L}\left(t,\theta,\beta,\dot{\theta},\dot{\beta}\right), \tag{49}$$

in which the parameter

$$\mathcal{L}=\tilde{J}_2^2\left(\sqrt{3}\sin 2\theta\cos\beta-3\cos^2\beta\sin^2\theta+3+\sin^2\theta\right)+4\tilde{J}_1^2\sin^2\theta, \tag{50}$$

and the couplings $\tilde{J}_{1,2}$ read

$$\tilde{J}_2=-\frac{\dot{\theta}}{2\sqrt{3}\sin\beta},\ \tilde{J}_1=\frac{\dot{\beta}}{4}-\frac{\dot{\theta}\cos\beta\cos\theta}{4\sin\beta\sin\theta}-\frac{\tilde{J}_2}{2}. \tag{51}$$

To make Q_r as small as possible, here we fix $\beta(t)$ with a linear function and express $\theta(t)$ as the modified sine Fourier series

$$\beta(t)=\frac{\pi}{2}\frac{t}{T}, \tag{52}$$

$$\theta(t)=\frac{2\pi}{3}\left(1-\frac{t}{T}\right)+\sum_n C_n\sin\left(\frac{2n\pi t}{T}\right). \tag{53}$$

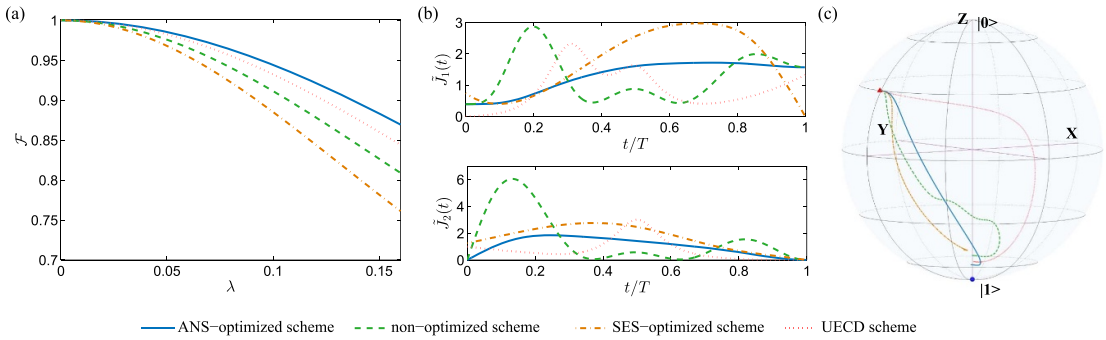


Figure 4. (a) The fidelity $\mathcal{F}$ versus the strength of amplitude noise λ using the ANS-optimized (blue solid line) with $C_n(n=1,2,3)=[0.2651,0.0279,0.0041]$, non-optimized scheme (green dashed line) with three random parameters being $C_n(n=1,2,3)=[-\frac{1}{6},\frac{1}{9},\frac{5}{54}]$, SES-optimized (orange dash-dotted line), and UECD (red dotted line) protocols, respectively. (b) The time-dependent couplings $\tilde{J}_{1,2}(t)$ for the protocols above-mentioned. (c) The corresponding evolutions on the Bloch sphere when $\lambda=0.1$, in which the red triangle and blue circle denote the initial and target state.

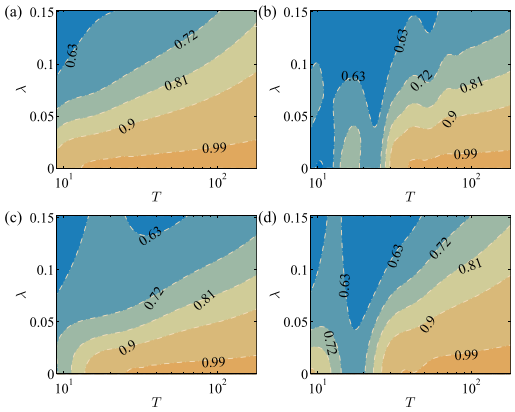


Figure 5. QST fidelity F in the five spin chain versus the amplitude noise parameter λ for different durations T using (a) the minimal ANS , (b) the non-optimized, (c) the optimized SES and (d) UECD schemes. The values of contour lines are labeled in each subplots. Here the inner couplings are set as $J_0=120$ MHz.

Here C_n is a set of to-be-optimized and time-independent coefficients, which should satisfy

$$\sum_n nC_n-\frac{1}{3}=0 \tag{54}$$

such that the boundary conditions $\dot{\theta}(t)\big|_{t=0,T}=0$ is guaranteed. In principle, the increasing number of C_n provides a better chance to obtain a smaller Q_r.

Figure 4(a) shows the fidelity $\mathcal{F}$ in the two-level effective system versus the noise parameter λ with the following shortcuts: (i) the ANS-optimized scheme with three coefficients, (ii) the not-optimized passage with three random coefficients in equation (53), (iii) the SES-optimized protocol, and (iv) the UECD driving. We can see that the ANS-optimized shortcut is more stable against the amplitude noise compared with the other methods. Figure 4(b) presents the time-dependent couplings $\tilde{J}_{1,2}(t)$ for the four schemes. Figure 4(c) shows the corresponding instantaneous trajectories on the Bloch sphere when $\lambda=0.1$ and $T=1$, from which we can see that the final state using optimized ANS scheme gets closer to the target state.

To further confirm the validity of our scheme, we substitute the original couplings $J_{1,2}(t)$ with the ANS-optimized ones above in the five-spin system, and we numerically investigate the spin-state transfer fidelity F, described by equation (37), with respect to the amplitude noise parameter λ when $J_0=120$ MHz. As displayed in figure 5, the final fidelity F in all speeding-up schemes reasonably descend with λ increasing, but the ANS-optimized scheme holds stronger robustness compared to other schemes. Though the optimal SES protocol is robust concerning systematic errors, it is sensitive to amplitude noise, showing the ultimate robustness depends on the specific type of perturbation. Notably, when T approaches zero, F in all cases obviously drop down, as the increasing couplings $J_{1,2}(t)$ invalidate the two-level approximate model. In principle, we can increase the inner couplings strength J_0 to mitigate this problem. However, it is intractable as the inner coupling of spin chain is finite in experiments. For example, the maximum strength of couplings

in typical quantum dots is usually about 100 MHz [28]. A higher-order approximation [54] should be introduced into the effective Hamiltonian. We will not discuss here the effects of higher-order approximation in equation (3), which is also an interesting issue but quite separate from our focus.

4. Conclusion

Faithful transfer of quantum state is necessary for quantum information processing. However, the inevitable environmental noise and imperfect system parameters will result in control fields deviating from ideal values, and hence the QST fidelity is significantly affected. Here we propose feasible, robust and high-fidelity STA schemes to transport quantum states along a Heisenberg spin chain with different types of perturbations being considered. To be specific, in the low-energy subspaces of an odd-size and strong-coupled spin channel, we approximate the Hamiltonian of the original system to a two-level system. Then, combining the perturbation theory and invariant-based engineering, we analyze and optimize the QST sensitivities with respect to the systematic errors and stochastic noise arising from the time-dependent couplings, respectively. Both protocols only require controlling two marginal couplings, which can greatly reduce the difficulty of implementations in experiments. Numerical simulations show that our schemes can provide robust and fast strategies for long-distance quantum information transport in various spin-based systems.

We would like to emphasize that our studies primarily focus on the analysis of systematic errors and amplitude noise in disordered couplings individually. In fact, both types of errors may take place together, and the optimal schemes in this situation usually depend on the ratio between amplitude–noise error and systematic error [56]. In addition, here we consider the special cases of white noise existing in the amplitudes of the disordered couplings. More complex types of noise [59], such as the Gaussian colored noise and non-Gaussian cases, can also be analyzed in principle. For instance, the influence of Ornstein–Uhlenbeck noise and flicker noise using STA for lattice transport have been studied [61, 62]. Combining the invariant-based engineering with optimal control techniques [52, 63, 64] will allow for better stability under different physical constraints and noise types during QST process, which is of course an important topic but beyond the scope of the present paper.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Appendix A. Unitarily equivalent CD driving

Counterdiabatic (CD) driving provides a simple way to exactly suppress nonadiabatic transitions by adding a CD term to the original Hamiltonian. For a general two-level system

$$\mathcal{H}_0 = \Gamma(t)\sigma^z + \omega(t)\sigma^x, \tag{A1}$$

the CD field can be calculated explicitly and it has the form $\mathcal{H}_{\rm cd}(t) = f_{\rm cd}(t)\sigma^y$ with

$$f_{\rm cd}(t) = \frac{\dot{\omega}\Gamma - \dot{\Gamma}\omega}{\omega^2 + \Gamma^2}. \tag{A2}$$

Despite taking a simple form in the two-level system, this CD term $\mathcal{H}_{\rm cd}(t)$ typically involves highly nonlocal many-body operators in the full Hilbert space [65, 66]. An alternative to construct feasible shortcuts from existing CD protocols is to perform suitable unitary transformations [67, 68].

If we apply the unitary rotation $U_z(\theta) = \exp(-i\theta\sigma^z/2)$ to the CD Hamiltonian

$$\mathcal{H}_{\rm A} = \mathcal{H}_0 + \mathcal{H}_{\rm cd} \tag{A3}$$

by a time-dependent angle $\theta(t) = \arctan\frac{f_{\rm cd}}{\omega}$, the auxiliary operators $\mathcal{H}_{\rm cd}(t)$ can be removed and the Hamiltonian of the unitarily equivalent CD (UECD) driving becomes

$$\mathcal{H}_{\rm U} = \left[\Gamma(t) - \frac{\dot{\theta}}{2}\right]\sigma^z + \left[\omega(t)\cos\theta + f_{\rm cd}(t)\sin\theta\right]\sigma^x, \tag{A4}$$

which is structurally equivalent to the original one in equation (A1). The wave functions and Hamiltonians in the two reference frames will coincide at the boundary times if $\mathcal{U}(0) = \mathcal{U}(T) = 1$ and $\dot{\mathcal{U}}(0) = \dot{\mathcal{U}}(T) = 0$. According to the equations (4) and (11), we can demand the original $J_{1,2}(t)$ follow the boundary constraints

$$\dot{J}_{1,2}(0) = 0, \quad \ddot{J}_{1,2}(T) = 0. \tag{A5}$$

Since the designed $\mathcal{H}_{\rm U}$ does not introduce new operators, nonexistent in $\mathcal{H}_0$, we only need to manipulate the qubit-chain couplings, which facilitate the experimental realization. Then, based on equation (A5), different driving protocols can be designed. In the main text, we choose the original couplings $J_{1,2}(t)$ as

$$J_1(t) = \frac{t^3}{T^3}, \quad J_2(t) = \left(1 - \frac{t}{T}\right)^3, \tag{A6}$$

and the redesigned couplings (red dotted lines) are presented in figure 4(b).

Appendix B. The performance with regard to different system sizes and the validity of weak-coupling condition

For a given evolution time T, the marginal couplings become stronger with increasing lengths of the spin chain, and they also have an impact on the final results. Figures 6(a) and (b) numerically present the QST fidelity as the length of the spin chain increases, where we take either the systematic deviations or stochastic amplitude noise into consideration, respectively. The lines without symbols denote the optimized schemes, and the corresponding lines with symbols represent the non-optimized results. The results show that as the system size N increases, the QST fidelity F still reaches a high level and exhibits good robustness against the perturbations. Note that for the ANS-optimized scheme, we only present the results up to an $N = 7$ spin chain limited by numerical computing abilities. The couplings $J_{1,2}(t)$ increase with the system size, which may invalidate the low-energy approximation. Therefore, the QST fidelity drops slightly for longer chains. In this situation, a stronger internal couplings J_0 contributes to a higher fidelity in principle. However, the strength of the spin–spin couplings is usually limited in experiments. To briefly illustrate the reasonable region of the internal coupling J_0 that ensures the validity of the weak-coupling condition, we perform numerical simulations in the $N = 7$ spin chain based on the SES-optimized scheme, without considering the perturbation. Figure 6(c) shows the final ground-state population $\mathcal{P}$ as a function of the relative proportion μ, between the maximum value of $J_{1,2}^{\max}$ and J_0, i.e.

$$\mu = \frac{\max\{J_1^{\max}, J_2^{\max}\}}{J_0}.$$

Here $J_{1,2}^{\max}$ denotes the maximum of $J_{1,2}(t)$ within the evolution time T. We find that the final ground-state population $\mathcal{P} > 0.99$ when $\mu < 0.2$. However, in the region $\mu > 0.2$, where the internal coupling is not strong enough, the population decreases obviously, which indicates the weak-coupling condition breaks down. If so, we need to make use of other methods for fast quantum-state transport.

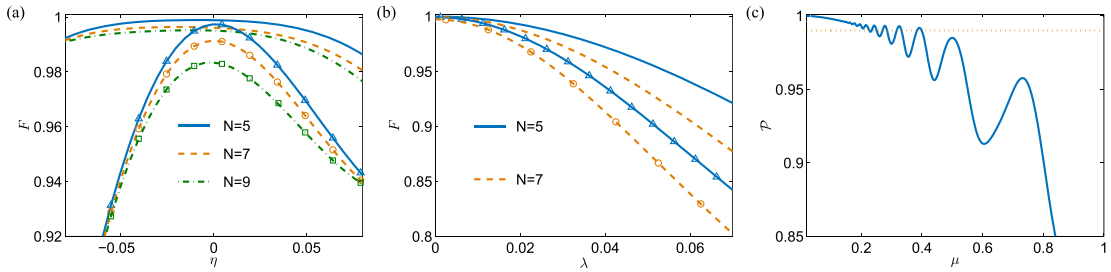


Figure 6. (a) and (b) The QST fidelity F versus the different lengths of the chain using the (a) SES-optimized scheme for $\mathcal{K}=1$ and (b) ANS-optimized protocol, respectively. In both cases the inner couplings $J_0 = 200$ MHz and evolution time $T = 60$ ns. (c) The final ground-state population $\mathcal{P}$ vs the parameter μ in the $N = 7$ spin chain using the SES-optimized scheme. The dotted line denotes a population of 0.99.

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